AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

MECHATRONICS ENGINEERING DEPARTMENT

PROJECT REPORT

PREPARED BY

Logien Abdelrahman	24p0391
Malak Marwan	24p0435
Isen Mohammed	24p0128
Tasabeeh Mohammed	24p0461
Youssef Atef Zaki	22p0148

Introduction

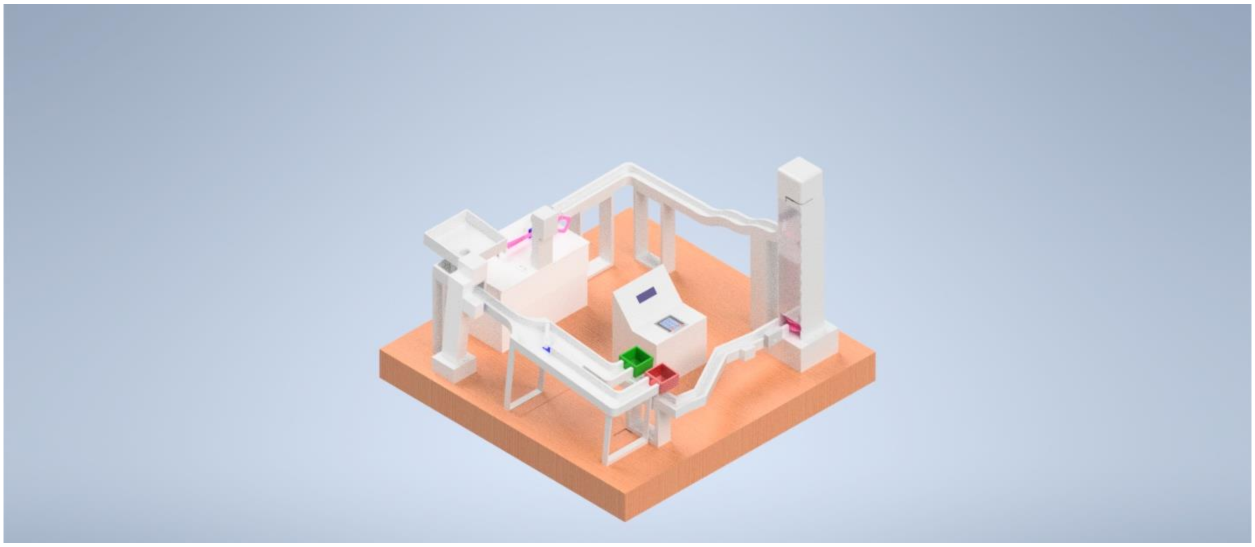
This report details the design and implementation of an interactive educational mechatronics system developed. The project aims to bridge the gap between theoretical mathematical concepts and physical mechanical action through an automated sorting and delivery mechanism.

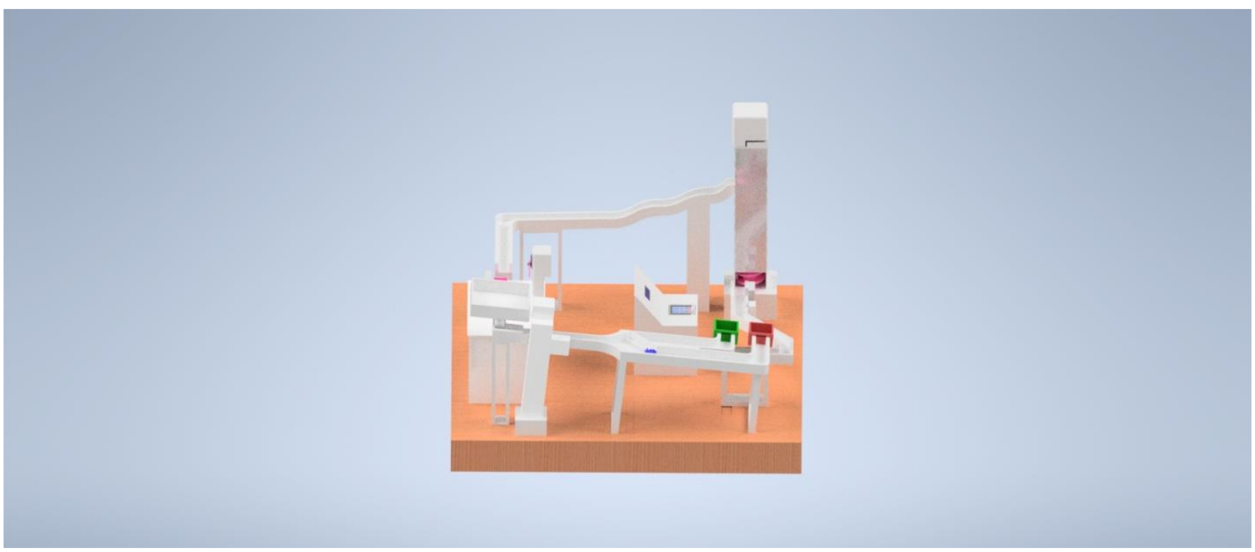
Project Overview and Concept

The core objective of this system is to create an engaging learning environment for children. The operational logic follows a feedback-controlled sequence:

1. User Interaction: A mathematical question is displayed on a digital interface.
2. Conditional Logic: If the child provides the correct answer, the system triggers the primary actuator to release a ball into the processing mechanisms.
    - If the answer **is incorrect,** the system remains in a "hold" state, and the ball does not move, encouraging the user to try again.
3. Mechanical Execution: Once a correct answer is detected, the ball travels through a series of integrated subsystems, including vertical lifting, sorting mechanisms, and a final projectile shooting phase.

Full CAD design





Design Process and Development Steps

1. Project Objective

The main objective of this project is to develop an automated ball sorting and handling system using Arduino-based control. The system combines color detection, object detection, user interaction, motor control, and servo mechanisms to create a complete automated process. The project was designed and simulated using Proteus before hardware implementation.

2. System Analysis

The required system functions were analyzed and divided into several stages:

1. User interaction and game control.
2. Ball detection.
3. Ball lifting mechanism.
4. Color recognition and sorting.
5. Ball shooting mechanism.
6. Communication between subsystems.

To simplify the design and reduce processing load, the system was divided into two Arduino controllers.

3. Hardware Selection

The following components were selected:

- Arduino Uno (2 units)
- LCD 16×2 I2C Display

- 4×3 Keypad
- HC-SR04 Ultrasonic Sensors
- Servo Motors
- DC Motor
- L298N Motor Driver
- TCS3200 Color Sensor
- Push Button
- Power Supply

4. System Architecture

The system was divided into two main subsystems:

Arduino 1

Responsible for:

- Color detection.
- Ball sorting mechanism.
- Shooting mechanism.
- Communication with Arduino #2.

Arduino 2

Responsible for:

- User interface.
- Mathematical Games.

- LCD display.
- Keypad input.
- Ball lifting mechanism.
- Motor control.
- Communication with Arduino 1.

5. Software Development

The software was developed using Arduino IDE.

The development process followed these stages:

Stage 1: LCD and Keypad Interface

The LCD was programmed to display instructions and mathematical questions. The keypad was used to receive user answers.

Stage 2: Mathematical Game Logic

Random arithmetic questions were generated. The user's answer was verified before allowing the next stage of operation.

Stage 3: Servo Gate Control

A push button was used to control the servo gate position. Debouncing techniques were implemented to ensure stable operation.

Stage 4: Ball Detection

The HC-SR04 ultrasonic sensor was used to detect the presence of a ball before activating the lifting mechanism.

Stage 5: Motor Control

The DC motor was controlled through an L298N motor driver. The motor operates for a predefined period to move the ball through the system.

Stage 6: Color Detection

The TCS3200 color sensor measures RGB values and compares them with calibrated reference values to identify ball color.

Stage 7: Sorting Mechanism

According to the detected color, servo motors move the sorting gate to direct the ball toward the appropriate path.

Stage 8: Shooting Mechanism

Additional servo motors perform the shooting sequence after successful sorting.

Stage 9: Arduino-to-Arduino Communication

Software Serial communication was implemented between the two Arduino boards to exchange system status messages such as:

- SHOOTING
- SORTING
- LIFTING

These messages are displayed on the LCD to provide user feedback.

6. Proteus Simulation

The simulation process was carried out in several steps:

1. Individual testing of each component.
2. Verification of sensor readings.
3. Verification of servo movements.
4. Testing motor driver functionality.
5. Testing LCD and keypad interaction.
6. Testing serial communication between both Arduino boards.
7. Integration of all subsystems into a complete model.
8. Final verification of system operation.

7. System Operation Sequence

The complete operating sequence is:

1. Users start the system using the keypad.
2. A mathematical question is displayed.
3. User enters the correct answer.
4. Ball is inserted into the system.
5. Ultrasonic sensor detects the ball.
6. Lifting mechanism transports the ball.
7. Color sensor identifies the ball color.
8. Sorting servos direct the ball to the proper path.

9. Shooting mechanism launches the ball.
10. System returns to standby mode and waits for the next cycle.

Bill of materials

Item	Quantity	Description	Function
Arduino Uno R3	2	Microcontroller Board	Main System Control & Processing

Item	Quantity	Description	Function
L298 Motor Driver	1	Dual H-Bridge Module	Motor Speed & Direction Control
Servo Motor (MG996R)	4	High Torque Metal Gear	Sorting & Shooting Mechanisms
DC Motor (12V)	1	High Torque Geared Motor	Main Drive System
TCS3200 Sensor	1	Color Recognition Sensor	Ball Detection & Sorting
Ultrasonic Sensor	2	HC-SR04	Distance & Obstacle Detection
I2C LCD Display	1	16x2 Character Display	User Interface & Status Monitoring
Keypad (4x4)	1	Membrane Keypad	User Input & Control

Item	Quantity	Description	Function
Power Supply	1	12V 5A Adapter	Main System Power
Jumper Wires	1 set	Male-to-Female/Male-to-Male	Circuit Interconnections
Breadboard	2	400-Point	Prototyping Connections

Work breakdown structure

1. Project planning and concept development

1.1 User Research

1.1.1 Observation & interviews

1.1.2 Identify emotional impact

1.1.3 Define educational benefit

1.2 Problem Definition

1.2.1 Requirement Analysis (sorting, shooting, lifting)

1.2.2 Define educational objectives

1.2.3 Define constraints (size, cost, safety)

1.3 Ideation

1.3.1Mind maps

1.3.2 Mood boards

1.3.3 Sketches

1.3.4 Concept selection

1.4 Product Design Specifications

2. Mechanical Subsystems

2.1 Vertical Lifting Mechanism

2.1.1 Mechanism selection

2.1.2 Torque calculation

2.1.3 Motor sizing

2.1.4 CAD modeling

2.1.5 Mechanical testing

2.2 Sorting Mechanism

2.2.1 Sorting principle (Color)

2.2.2 Sensor selection

2.2.3 Mechanical gate design (inventor)

2.2.4 Sorting validation testing

2.3 Projectile Shooting Mechanism

2.3.1 Concept selection (catapult)

7.4.1 Wiring per Module

7.4.2 Cable management

7.5 System Integration

7.5.1 Mechanical–Electrical Integration

7.5.1.1 1Install motors into mechanical structure

7.5.1.2 Mount sensors in final positions

7.5.1.3 Connect actuators to control system

7.5.2 Electrical–Control Integration

7.5.2.1 Connect drivers to microcontrollers

7.5.2.2 Upload Code to MCU1

7.5.2.3 Upload code to MCU2

7.5.2.4 Establish communication protocol between MCUs

7.5.3 User Interface Integration

7.5.3.1Connect LCD to main controller

7.5.3.2 Implement UI logic

7.5.3.3 Integrate Bluetooth module

7.5.3.4 Connect Processing app to MCU

7.5.4 Power System Integration

7.5.4.1 Connect single power source

7.5.5 Module Interconnection

7.5.5.1 Connect modular wiring plugs

7.5.5.2 Final assembly to main board

8. Testing and Validation

8.1 Mechanical testing

8.1.1 Screw efficiency test

8.1.2 Sorting accuracy test

8.1.3 Launch consistency test

8.2 Electrical Testing

8.2.1 power stability

8.2.2 sensor accuracy

8.2.3 motor control

8.3 system integration testing

8.3.1 Mechanical–Electrical Integration

8.3.1.1 Verify motors respond to control signals

8.3.1.2 Verify sensors trigger mechanical actions

8.3.2 MCU Communication Verification

8.3.2.1Test MCU1 ↔ MCU2 communication

8.3.2.2 Validate data transmission

8.3.3 Closed-Loop Operation Test

8.3.3.1 Marble insertion test

8.3.3.2Full internal cycle validation

8.3.3.3 Marble retention check (no escape from system)

8.3.4 User Interface Integration

8.3.4.1 LCD displays correct real-time data

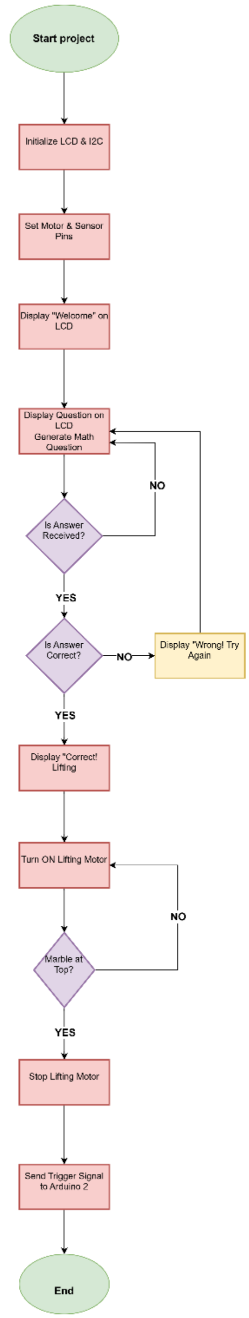
8.3.4.2 App updates correctly

8.3.4.3 Button inputs trigger correct actions

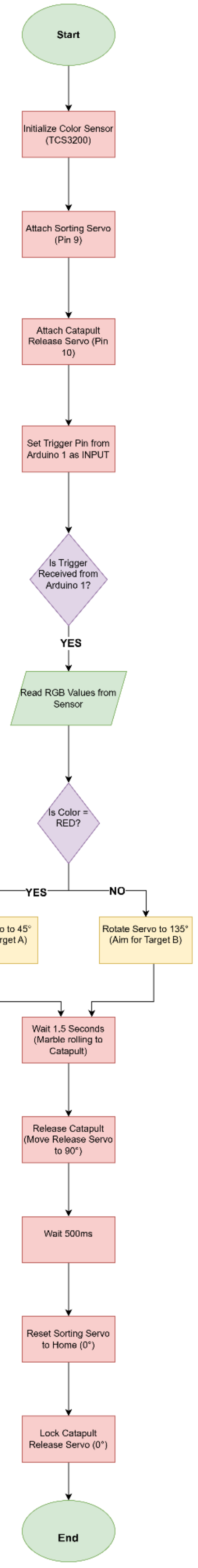
9. documentation and deliverables

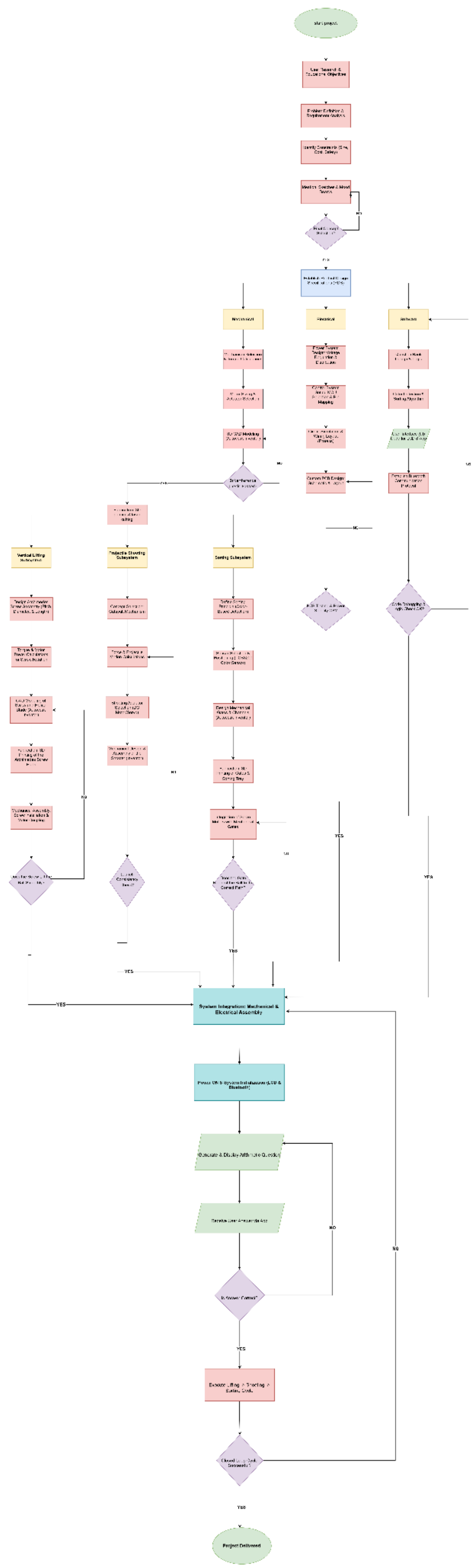
9.1 Technical Report

Flowchart

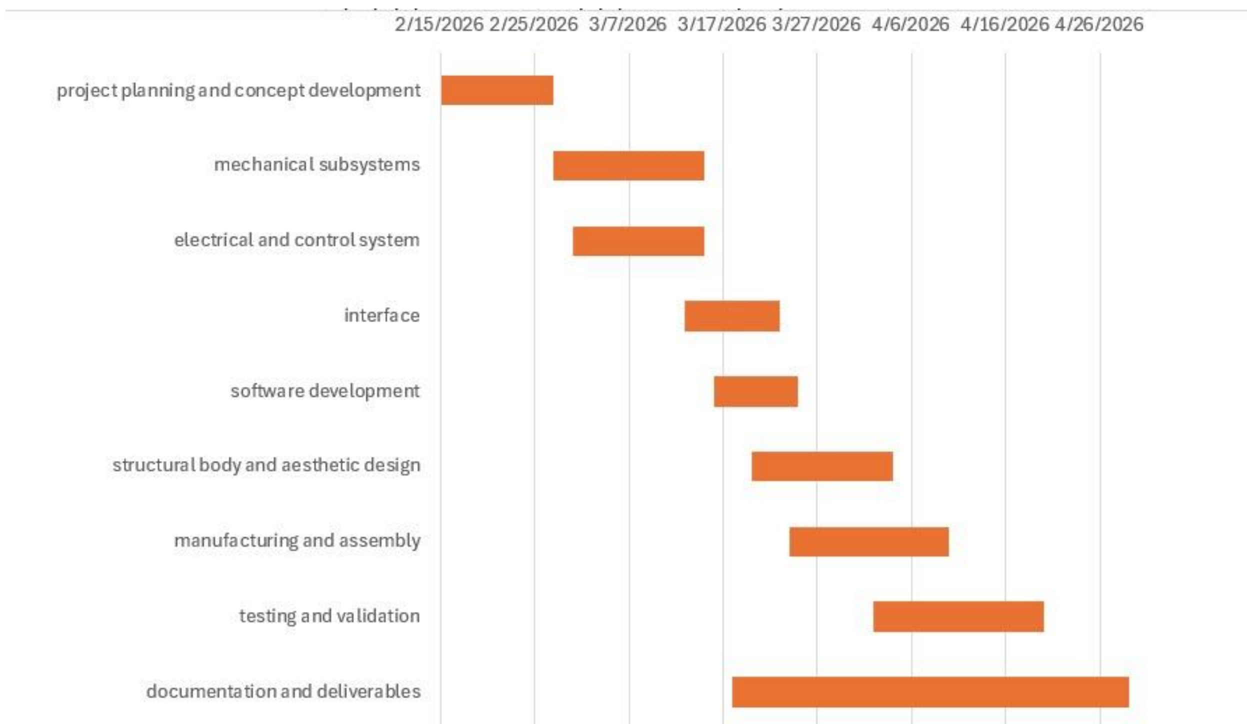


Start project
Initialize LCD & I2C
Set Motor & Sensor Pins
Display "Welcome" on LCD
Display Question on LCD
Generate Math Question
Is Answer Received?
NO
YES
Is Answer Correct?
NO
Display "Wrong! Try Again
YES
Display "Correct! Lifting
Turn ON Lifting Motor
Marble at Top?
NO
YES
Stop Lifting Motor
Send Trigger Signal to Arduino 2
End


Start
Initialize Color Sensor (TCS3200)
Attach Sorting Servo (Pin 9)
Attach Catapult Release Servo (Pin 10)
Set Trigger Pin from Arduino 1 as INPUT
Is Trigger Received from Arduino 1?
YES
Read RGB Values from Sensor
Is Color = RED?
YES
NO
Rotate Servo to 45° (Aim for Target A)
Rotate Servo to 135° (Aim for Target B)
Wait 1.5 Seconds (Marble rolling to Catapult)
Release Catapult (Move Release Servo to 90°)
Wait 500ms
Reset Sorting Servo to Home (0°)
Lock Catapult Release Servo (0°)
End


Vertical Lifting Subsystem
Projectile Shooting Subsystem
Sorting Subsystem
YES
NO
System Integration: Mechanical & Electrical Assembly
YES
Project Delivered

Gantt Chart



Product design specifications

Product overview:

The product shall be an interactive educational toy designed to support children with dyslexia through structured multisensory learning. The system integrates mechanical subsystems (lifting, sorting and shooting) with electronic feedback to create an engaging cause -and-effect learning environment.

Requirement category	Specifications
Target audience	• Children aged 6-10 years • Diagnosed or showing symptoms of dyslexia • Usable in homes, therapy centers, and schools • Parents struggling with their dyslexic children
User requirements & Research	• **User Needs:** Specific needs identified through user research. • **Emotional Impact**: The intended emotional response or engagement for the child. • **Educational Impact**: Description of how the toy aids in learning or skill development.
Mechanical requirements	• **Vertical Lifting Mechanism:** This system is responsible for moving the marbles from a lower reservoir or entry point to a higher elevation to gain potential energy. • **Sorting Mechanism:** The toy must be able to automatically categorize marbles based on physical property (color). • **Projectile-Based Shooting Mechanism:** After the lifting and sorting phases, the toy must launch the marbles toward a target using a mechanism like a catapult or a spring-loaded launcher.
System flow (closed-loop operation)	The toy is required to operate as a self-contained environment: • **Single Entry Point:** Users should insert marbles through one designated inlet.

	• **Internal Pathway:** Once inserted, the marble must stay within the toy's internal tracks or mechanisms throughout the entire operation cycle. • **Continuous Loop:** The flow should ideally transition from entry to lifting, then sorting, and finally shooting, potentially returning the marble to the start or a collection bin within the 100 cm X 100 cm footprint.
Control electronics	This section details the "brain" and power architecture of your mechatronic system: • **Dual Microcontroller Setup:** At least two microcontrollers (such as the **Arduino Uno**) that communicate with each other. • **Custom PCB Requirements:** At least one part of the circuit must be built on a custom-manufactured PCB rather than just a breadboard or protoboard. • **Single power source:** The entire system including all motors, sensors, and screens must be powered by a single unified power supply. • **Dual User Interfaces:** 1. Interface 1: An LCD connected directly to the Arduino to show real-time data. 2. Interface 2: A mobile or PC application created with Processing.io that communicates with the toy via Bluetooth.
Manufacturing & Maintenance	• **Modularity:** Requirements for individual wiring plugs and assembly connections for easy disassembly. • **Materials:** List of prototyping tools and materials suitable for a child's toy.

	• **Technical Documentation:** Requirements for Autodesk Inventor (mechanical) and Proteus (electrical) files.
Modularity and Safety	• **Modular Assembly:** The project must be divided into modules, each with its own wiring and connection plugs, to allow for easy assembly and disassembly. • **Hidden Wiring:** To ensure safety for the 2–10 age group, all wires and electrical components must be hidden from view.
Power consumption	• This section details the power architecture of the mechatronic system, ensuring compliance with the "Single power source" requirement and the safety standards for the target audience (children aged 6-10).

For lifting mechanism:

Component	Voltage	Current	Power

Dc motor	12 V	0.6 A	7.2 W
2 IR sensors	5 V	0.04 A	0.2 W
Total	12 V	0.64 A	7.68 W

Total time = 7.5 sec

For sorting mechanism:

Component	**Voltage**	**Current**	**Power**
Color sensor (TCS3200)	5 V	0.02 A	0.1 W
Sorting servo motor	5 V	0.5 A	2.5 W
Total	5 V	0.52 A	2.6 W

Total time = 0.9 sec

For shooting mechanism:

Component	Voltage	Current	Power
High torque servo (catapult)	5 V	1 A	5 W
Total	5 V	1 A	5 W

Total time: 0.5 sec

Sure. You can copy the following section directly into your report:

Subsystem Current Consumption

Component	Quantity	Current per Unit (mA)	Total Current (mA)
Arduino Uno	2	50	100
LCD 16×2 I2C	1	30	30
Keypad 4×3	1	1	1
HC-SR04 Ultrasonic Sensor	2	15	30
TCS3200 Color Sensor	1	20	20
Servo Motors	4	500	2000
DC Motor	1	800	800
L298N Motor Driver	1	20	20

Total Current Consumption

The total current consumption of the system is calculated as follows:

I total = 100 + 30 + 1 + 30 + 20 + 2000 + 800 + 20

I total = 3001 mA

I total ≈ 3.0 A

Safety Factor

To account for startup currents, load variations, and future expansion, a safety factor of 25% is added:

I required = 1.25 × 3.0

I required = 3.75 A

Power Supply Selection

Based on the calculated current requirements, a power supply capable of delivering at least 4 A is recommended to ensure stable and reliable operation of the complete mechatronic system.

Therefore, the recommended power supply specifications are:

- Control Voltage: 5 V DC
- Minimum Current Rating: 4 A
- Motor Supply Voltage: According to the DC motor specifications

This power supply provides sufficient current for all sensors, actuators, controllers, and communication modules while maintaining safe operating conditions.

Battery Sizing

The battery capacity must be sufficient to supply the required current for the desired operating time of the mechatronic system.

Total Current Requirement

From the current consumption analysis, the total system current is approximately:

I total = 3 A

Including a safety margin of 25%:

I required = 3.75 A

Battery Capacity Calculation

Battery capacity is calculated using:

Capacity (Ah) = Current (A) × Operating Time (h)

Assuming the system is required to operate continuously for 2 hours:

Capacity = 3.75 × 2

Capacity = 7.5 Ah

Battery Selection

The battery is selected with a capacity greater than the calculated value to compensate for efficiency losses and battery aging.

Recommended battery specifications:

- Battery Voltage: 12 V DC
- Battery Capacity: 8 Ah minimum
- Recommended Capacity: 10 Ah

Energy Calculation

The required battery energy can be estimated as:

Energy (Wh) = Voltage (V) × Capacity (Ah)

For a 12 V, 10 Ah battery:

Energy = 12 × 10

Energy = 120 Wh

Actuator sizing

For lifting mechanism:

Motion profile:

The motion is divided into three distinct functional phases to protect the motor's gearbox and maintain the stability of the load:

Acceleration Phase (t acceleration): The actuator ramps from rest to the maximum operational angular velocity (omega_{max}). This prevents sudden torque spikes and motor stalling.

- Duration: **0.5 s**.

Constant Velocity Phase (t constant): The screw rotates at a steady state, providing a stable window for the color and proximity sensors to sample data.

- Calculation: t constant = T total - (t acceleration + t deceleration)

  T constant = 7.5 – (0.5 + 0.5) = 6.5 s

- Duration: **6.5 s**

Deceleration Phase (t deceleration): The velocity is decreased linearly to zero. This ensures the ball is positioned accurately for the projectile mechanism without overshooting due to inertia.

- Duration: **0.5s**

System Synchronization and Dwell Time: The Dwell Time is defined as the stationary period between two consecutive lifting cycles. In this specific mechanism, it serves as a synchronization buffer to ensure:

- Inter-mechanism Handover: The ball is successfully transferred from the Archimedes screw to the sorting/projectile assembly.
- Actuator Reset: Providing sufficient time for the projectile servos or solenoids to return to their "home" position.

- Sequential Logic: Preventing the next ball from entering the discharge zone before the previous one has been fully processed.
- T dwell = 2 sec

Kinematic Calculations

- Total Distance (L): 0.5 m.
- Pitch (S): 0.0262 m
- Total Time (T): 7.5 sec
- N = p / h = 0.0262 / 0.5 ≈ 19
- Total angular displacement:

Theta total = 2 pi N = 119.38 rad

- Maximum angular velocity:

W max = theta total / 7 = 119.38 / 7 = 17.05 rad / sec

- Angular acceleration:

Alpha = W max / ta = 17.05 / 0.5 = 34.1 rad / sec^2

For acceleration phase:

Theta 1 = ½ x W max x ta = ½ x 17.05 x 0.5 = 4.2625 rad

For constant phase:

Theta 2 = W max x tc = 17.05 x 6.5 = 110.825 rad

For deceleration phase:

Theta 3 = 4.2625 rad

- Theta total = 119.35 rad

Inertia:

The motor rotor inertia (J motor) was estimated based on the standard inertia matching principle for small-scale mechatronic systems. It is assumed to be approximately 10% of the load inertia (J load).

J screw = ½ (m screw) r^2 = ½ . 0.897 . (32.5 x (10) ^-3)^2 = 4.73728 x 10^-4 (kg.m^2)

J motor = 10% J screw = 4.73728 x 10^-5 (kg.m^2)

Total system inertia = j load + j motor = 5.211 x 10^-4 (kg.m^2)

Efficiency = 40% = 0.4

F = mg = 0.004 x 9.81 = 0.03924 N

Torque during motion phases:

Lead torque:

Tl = Fl / (efficiency x 2 pi pitch) = 0.03924 / (0.4 x 2 x pi x 0.0262) = 0.5959 N.m

Torque motor:

Torque due to acceleration = (j total x angular acceleration) + Tl = (5.211 x 10^-4 x 34.1) + 0.5959 = 0.614 N.m

Constant speed torque = (j total x angular acceleration) + Tl = 0.5959 N.m

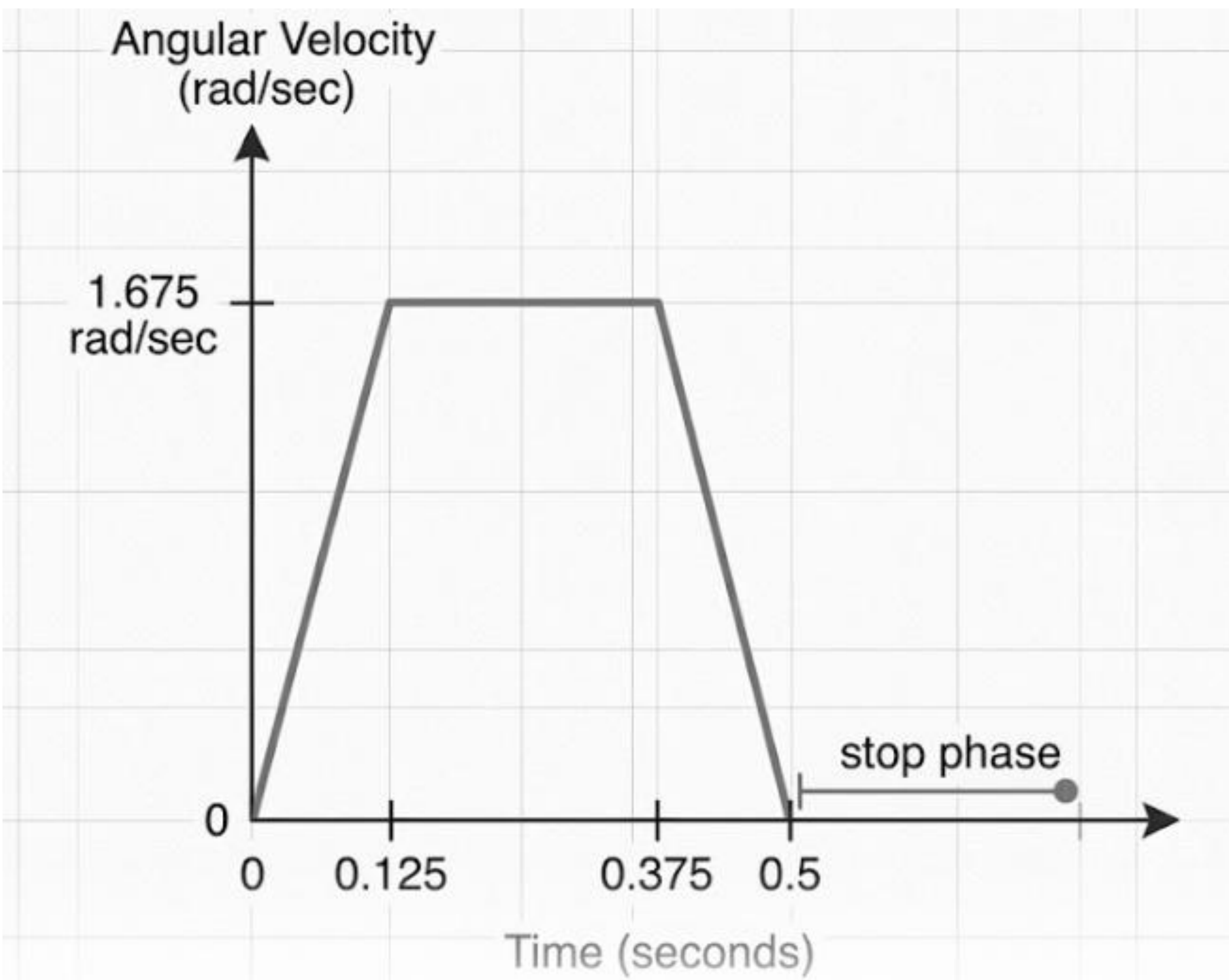
Torque due to deceleration = - (j total x angular acceleration) + Tl = $-(5.211 \times 10^{-4} \times 34.1) + 0.5959 = 0.578$ N.m

RMS torque = $\mathrm{sqrt}((T_{acc}^2 \times 0.5 + T_{const}^2 \times 6.5 + T_{dec}^2 \times 0.5) / (7.5 + 2)) = 0.53$ N.m

Using safety factor = 1.5

T maximum = $1.5 \times 0.614 = 0.921$ N.m

T RMS = $1.5 \times 0.53 = 0.8$ N.m



For shooting mechanism:

Motion profile

A motion profile describes how position, velocity, and acceleration of a system change over time during movement. In actuator sizing, it is used to determine the required torque and speed needed to achieve smooth and efficient operation of the mechanism.

Angular displacement $\Delta\theta=60^{\circ}=1.047$ rad.

Total motion time = 0.5 sec

Acceleration Phase (t acceleration):

The acceleration phase is the part of the motion profile where the actuator increases its speed from rest to the desired velocity. During this phase, the required torque is highest because the system must overcome both inertia and external load in the catapult.

From 0 to W maximum

- Duration: 0.125 sec

Constant Velocity Phase (t constant):

The constant velocity phase is the part of the motion profile where the actuator moves at a steady speed with zero acceleration. During this phase, the required torque is mainly used to overcome external load since inertia effects are minimal.

- Duration: 0.25 sec

Deceleration Phase (t deceleration):

The deceleration phase is the part of the motion profile where the actuator reduces its speed from the set velocity back to rest. During this phase, torque is applied in the opposite direction to control and stop the motion smoothly while overcoming the system's inertia and external loads.

From W maximum to 0

- Duration: 0.125 sec

System Synchronization and Dwell Time:

System synchronization and dwell time refer to the coordination of different parts of the system and the intentional pause between motions. Dwell time allows the mechanism to remain in a fixed position for a short period, ensuring proper timing between actions such as loading, holding, and releasing in the catapult system.

- T dwell = 9 sec

Kinematic calculations:

Θ = (½ x α x ta^2) + ((α ta) t constant) + (½ x α x td^2)

Δθ = α x ta^2

$$1.047 = \alpha(0.015625)$$

$$\alpha = \frac{1.047}{0.03} = 13.4 \text{ rad/sec^2}$$

$$1.047 = 7.8125\ x\ 10^{-3}\alpha + 0.0625\ \alpha + 7.8125\ x\ 10^{-3}\alpha$$

$$1.047 = 0.078\ \alpha$$

$$\alpha = 13.4\ \text{rad/sec\^{}2}$$

Maximum angular velocity = W maximum = alpha x ta = 13.4 x 0.125 = 1.675 rad / sec

For acceleration phase:

Theta 1 = ½ x alpha x ta^2 = ½ x 13.4 x (0.125) ^2 = 0.1047 rad

For constant velocity phase:

Theta 2= W max x tc = (1.675) (0.25) = 0.41875 rad

For deceleration phase:

Theta 3 = ½ x alpha x td^2 = ½ x 13.4 x (0.125) ^2 = 0.1047 rad

Θ total = 0.1047 + 0.41875 +0.1047 = 0.62815 rad

Inertia:

Given parameters:

Mass of the arm = 0.02 Kg

Length of the arm = 0.21 m

Mass of the ball = 0.004 Kg

distance from pivot to ball center $r = 0.21$ m

J arm = 1/3. m (arm). (L (arm)) ^2 = 1/3 x 0.02 x (0.21) ^2 = 2.94 x 10^-4 (kg.m^2)

J ball = m (ball) x r^2 = 0.004 x (0.21) ^2 = 1.764 x 10^-4 (kg.m^2)

Total inertia of the shooting arm assembly = J arm + J ball = 4.704 x 10 ^-4 (kg.m^2)

Torque during motion phases:

Torque during the motion phases varies according to the speed and acceleration of the mechanism. During acceleration, the actuator must provide the highest torque to overcome both inertia and external resistance, during constant velocity it mainly overcomes the load torque, and during deceleration it applies torque in the opposite direction to slow and stop the motion smoothly.

Calculations:

Rotation: $90° \rightarrow 20° \rightarrow \Delta\theta = 70° = 1.22173$ rad

Efficiency = 40 % = 0.4

Lead torque:

F dynamic = T / r = 6.303 x 10^-3 / 0.21 = 0.030016 N

F lead = F dynamic / efficiency = 0.030016 / 0.4 = 0.07504 N

Tl = (Fl x r) / efficiency = (0.07504 x 0.21) / 0.4 = 0.04 N.m

Elastic torque = F (force from the elastic band) x r (distance from the pivot to where the band is attached) = 5 x 0.21 = 1.05 N.m

Torque motor:

Torque due to acceleration = (j total x angular acceleration) + Tl = ($4.704 \times 10^{-4} \times 13.4$) + 0.04 = 0.0463 N.m

Constant speed torque = (j total x angular acceleration) + Tl = Tl = 0.04 N.m

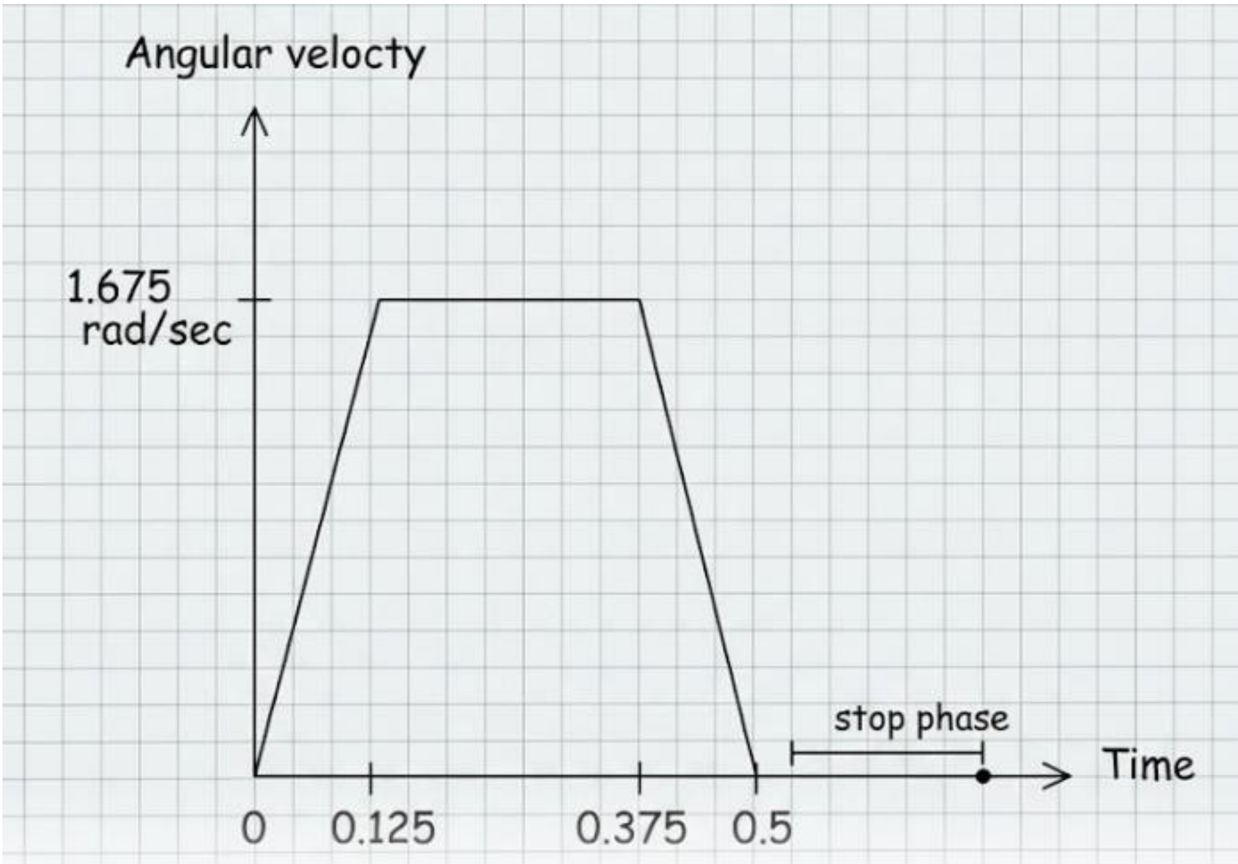
Torque due to deceleration = - (j total x angular acceleration) + Tl = -($4.704 \times 10^{-4} \times 13.4$) + 0.04 = 0.0337 N.m

RMS torque = sqrt (($T_{acc}^2 \times 0.125 + T_{const}^2 \times 0.25 + T_{dec}^2 \times 0.125$) / 0.5 + 9)) = 9.23×10^{-3} N.m

Using safety factor = 1.5

T max = 1.5 x 0.0463 = 0.06945 N.m

T RMS = 1.5 x 9.23×10^{-3} = 0.1385 N.m



Projectile calculations:

Procedure:

The projectile motion of the mechanism was initially analyzed by measuring the time of flight and approximating the launch angle in order to position the target basket. However, despite taking multiple readings and calculating the average, the measured time was found to be unreliable due to the very short duration of motion, which magnifies human reaction time error.

Therefore, a more accurate approach was adopted by positioning the target and calculating the required launch angle to reach it based on measured height and horizontal displacement.

Desired and Given values:

H (max height) = 0.22m

R (max horizontal distance) = 0.42 m

g= 9.81 m/s^2

1) Launch Angle

$$\boldsymbol{\theta = \tan^{-1}\left(\frac{4H}{R}\right)}$$

$$\boldsymbol{\theta = \tan^{-1}\left(\frac{4 \times 0.29}{0.42}\right)}$$

$$\boldsymbol{\theta = \tan^{-1}(2.762)}$$

$$\boxed{\boldsymbol{\theta = 70.1^\circ}}$$

2) Launch Velocity

$$\boldsymbol{v = \sqrt{\frac{Rg}{\sin 2\theta}}}$$

$$\boldsymbol{2\theta = 140.2^\circ, \sin 140.2^\circ \approx 0.643}$$

$$\boldsymbol{v = \sqrt{\frac{0.42 \times 9.81}{0.643}}}$$

$$\boldsymbol{v = \sqrt{6.41}}$$

$$\boxed{\boldsymbol{v = 2.53} \text{ m/s}}$$

3) Velocity Components

$$v_x = v\cos\theta = 2.53\cos 70.1^\circ$$

$$\boxed{v_x = 0.87 \text{ m/s}}$$

$$v_y = v\sin\theta = 2.53\sin 70.1^\circ$$

$$\boxed{v_y = 2.38 \text{ m/s}}$$

4) Time of Flight

$$T = \frac{2v\sin\theta}{g}$$

$$T = \frac{2(2.53)\sin 70.1^\circ}{9.81}$$

$$T = \frac{4.76}{9.81}$$

$$\boxed{T = 0.485 \text{ s}}$$

5) Maximum Height (Verification)

$$H = \frac{v^2\sin^2\theta}{2g}$$

$$H = \frac{(2.53)^2\sin^2 70.1^\circ}{2(9.81)}$$

$$\boxed{H \approx 0.29 \text{ m}}$$

6) Range (Verification)

$$\boldsymbol{R = \frac{v^2 \sin 2\theta}{g}}$$

$$\boldsymbol{R = \frac{(2.53)^2 \times 0.643}{9.81}}$$

$$\boxed{\boldsymbol{R \approx 0.42}\ \text{m}}$$

7) Time to Maximum Height

$$\boldsymbol{t_{max} = \frac{v_y}{g}}$$

$$\boldsymbol{t_{max} = \frac{2.38}{9.81}}$$

$$\boxed{\boldsymbol{t_{max} = 0.243}\ \text{s}}$$

8) Coordinates of Maximum Height

$$x_{max} = v_x \cdot t_{max} = 0.87 \times 0.243$$

$$\boxed{x_{max} \approx 0.21\ \text{m}}$$

$$\boxed{y_{max} = 0.29\ \text{m}}$$

9) Trajectory Equation

$$y(x) = x\tan\theta - \frac{gx^2}{2v^2\cos^2\theta}$$

$$\tan 70.1^\circ \approx 2.76, \cos^2 70.1^\circ \approx 0.116$$

$$y(x) = 2.76x - \frac{9.81x^2}{2(2.53)^2(0.116)}$$

$$y(x) = 2.76x - 6.57x^2$$

$$\boxed{y(x) = 2.76x - 6.57x^2}$$

For sorting mechanism:

Motion Profile

Motion Profile is a mathematical model that defines the position, velocity, and acceleration of an actuator over a specific duration. For a sorting mechanism integrated with a color sensor, the motion profile serves as the critical bridge between digital detection and physical execution. Once the color sensor identifies the marble, the motion profile dictates how the sorting arm or diverter transitions to the correct lane with speed and precision.

Acceleration Phase (t acceleration):

It is the period where an actuator transitions from a state of rest to its target operating speed.

From 0 to W maximum

- Duration: 0.3 sec

Constant Velocity Phase (t constant):

During this stage, the angular velocity remains stable, and the acceleration drops to zero (alpha = 0). This phase represents the steady movement of the diverter arm as it travels toward its final sorting position.

- Duration: 0.3 sec

Deceleration Phase (t deceleration):

This phase is just as critical as acceleration. It ensures that the actuator reaches its exact target position without mechanical impact or "overshoot."

- Duration: 0.3 sec

System Synchronization and Dwell Time:

- T dwell = 9 sec

Kinematic calculations:

Theta = 31 degrees = 0.541 rad.

- According to the area under the curve:

Delta theta = alpha x ta x (ta + tc) = 0.541 rad

$Alpha \times 0.3 \times 0.6 = 0.541$ rad

Angular acceleration = 3.0058 rad / sec^2

Maximum angular velocity = alpha x ta = 3.0058 x 0.3 = 0.9017 rad / sec

For acceleration phase:

Theta 1 = ½ x alpha x ta^2 = ½ x 3.0058 x (0.3)^2 = 0.135261 rad

For constant velocity phase:

Theta 2 = W max x tc = 0.9017 x 0.3 = 0.270522 rad

For deceleration phase:

Theta 3 = 0.135261 rad

Theta total = 0.135261+ 0.270522 + 0.135261 = 0.541044 rad.

Inertia

Given parameters:

Mass of the arm = 0.01 Kg

Length of the arm = 0.08 m

The sorting arm is a 3D-printed part rotating about a fixed pivot.

J arm = 1/3 x mass (arm) x l^2 = 1/3 x 0.01 x 0.08^2 = 2.13 x 10^-5 Kg.m^2.

J marble = M marble x R^2 = 0.004 x (0.06) ^2 = 1.44 x 10^-5 Kg.m^2.

(Where R is the distance from the pivot to the point where the marble hits the arm).

J total = J arm + J marble = 2.13 x 10^-5 + 1.44 x 10^-5 = 3.57 x 10^-5 Kg.m^2.

Torque during motion phases:

Torque (T) is the rotational effort required by the servo motor to move the sorting arm through its mission. Torque is not a single, static value; it is a dynamic requirement that changes based on the physics of each phase in the motion profile.

Dynamic torque = angular acceleration x total inertia = 3.0058 x 3.57 x 10^-5 = 1.071 x 10^-4 N.m

Lead torque = dynamic torque / efficiency = 1.071 x 10^-4 / 0.4 = 2.6775 x 10^-4 N.m

Torque motor:

Torque due to acceleration = (j total x angular acceleration) + Tl = (3.57 x 10^-5 x 3.0058) + 2.6775 x 10^-4 = 3.7485 x 10^-4 N.m

Constant speed torque = (j total x angular acceleration) + Tl = Tl = 2.6775 x 10^-4 N.m

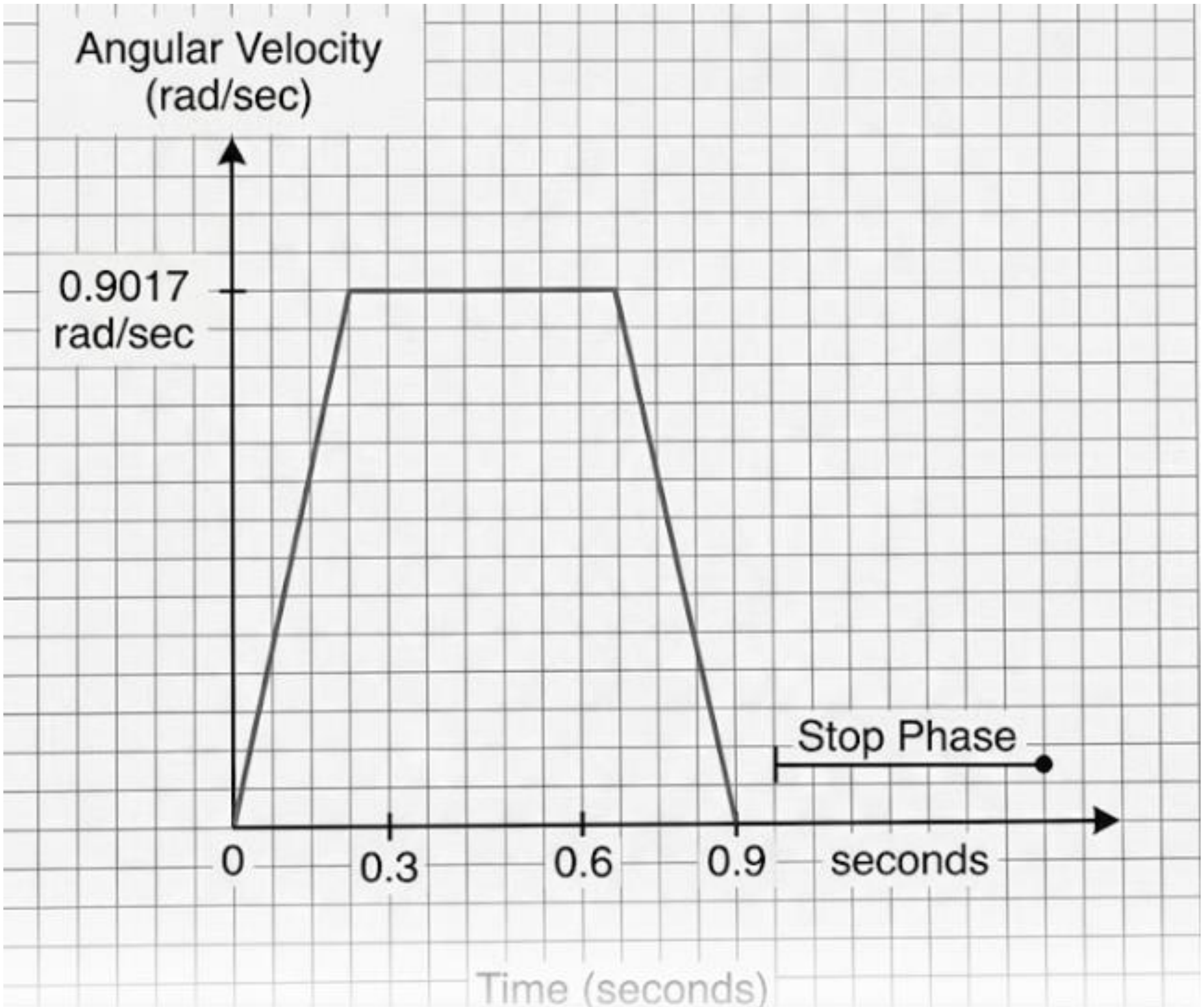
Torque due to deceleration = -(j total x angular acceleration) + Tl = = -(3.57 x 10^-5 x 3.0058) + 2.6775 x 10^-4 = 1.6065 x 10^-4 N.m

RMS torque = sqrt ((Tacc² × 0.3 + Tconst² × 0.3 + Tdec² × 0.3) / 0.9 + 9)) = 8.5 x 10^-5 N.m

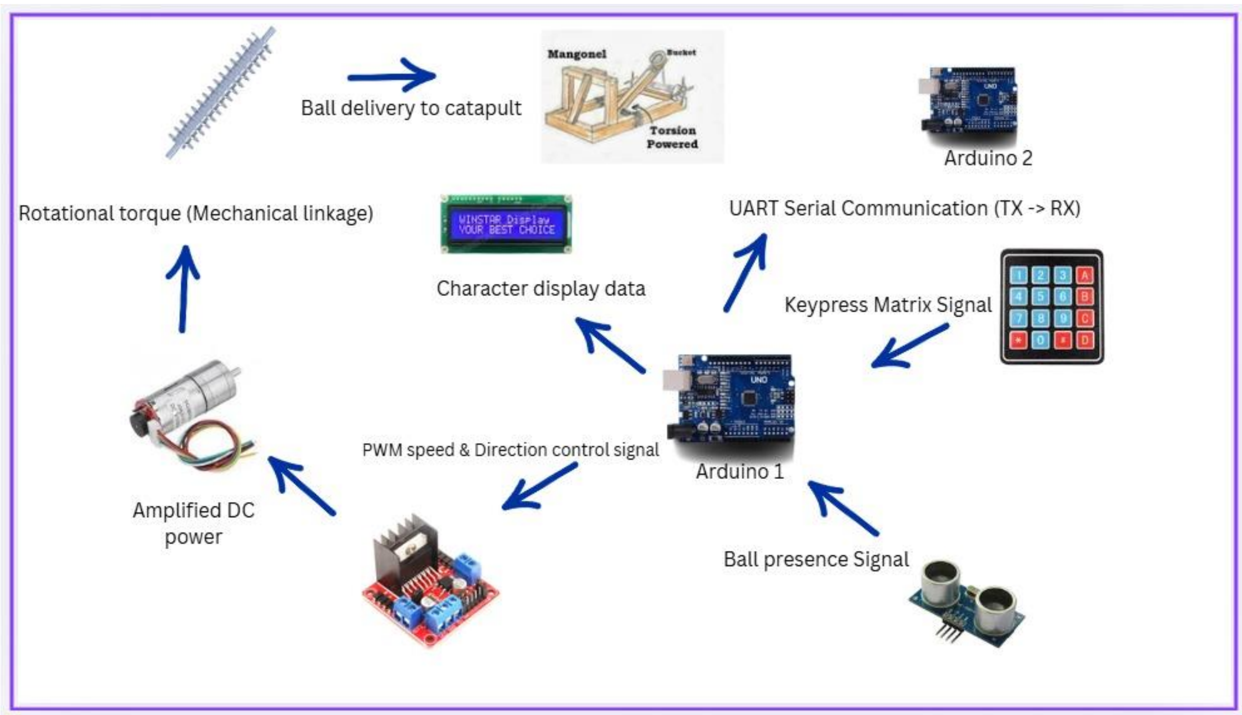
Using safety factor = 1.5

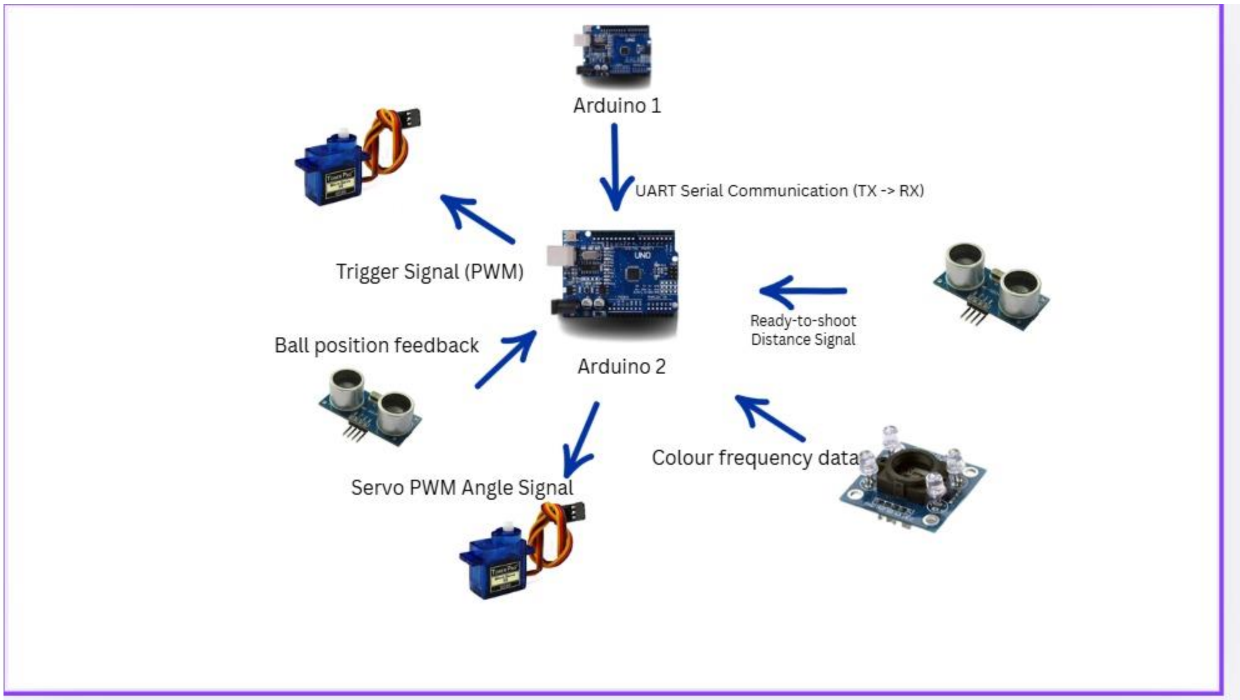
T max = 1.5 x 3.7485 x 10^-4 = 5.62275 x 10^-4 N.m

T RMS = 1.5 x 8.5 x 10 ^ -5 = 1.28 x 10^-4



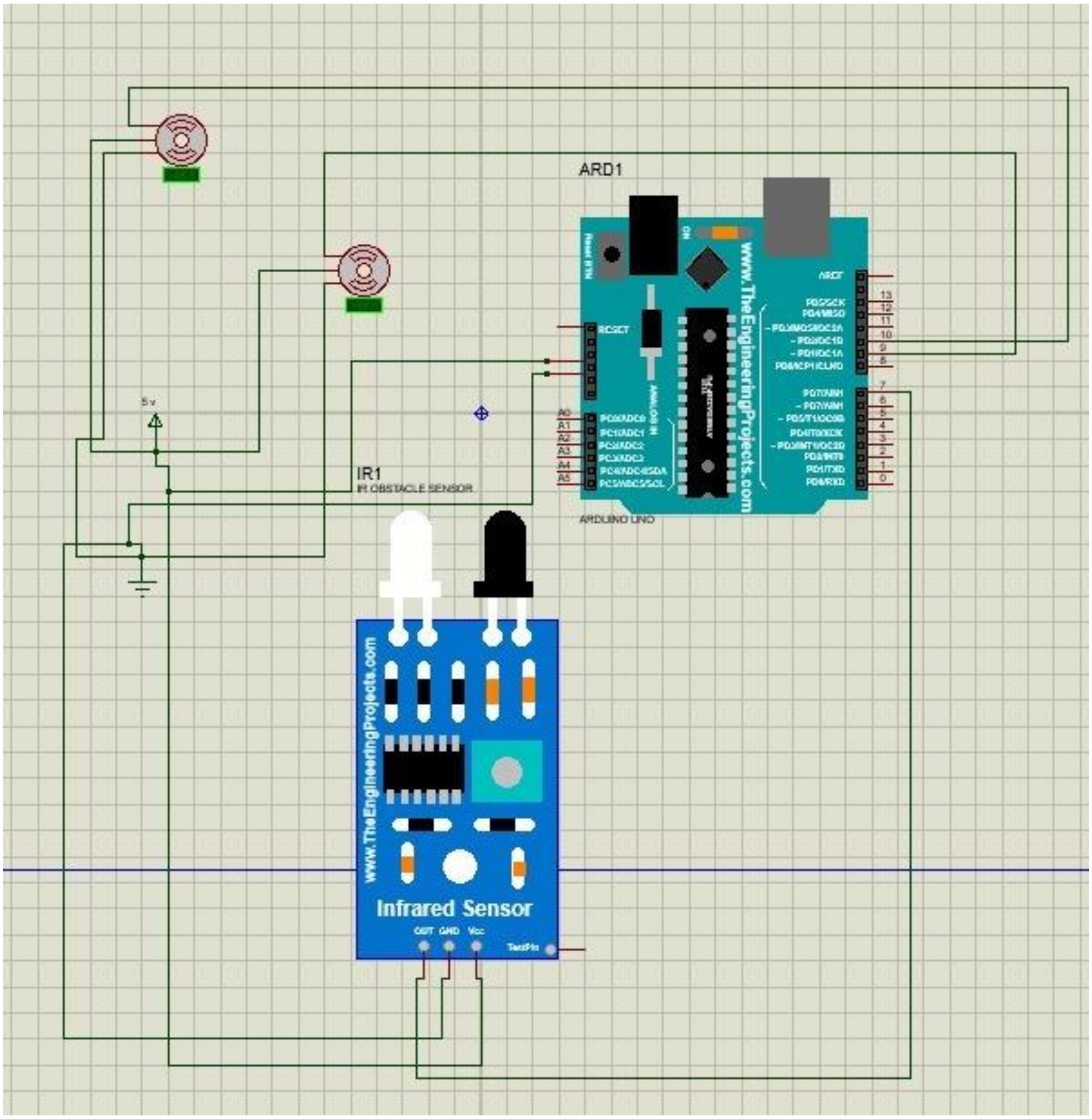
Block Diagram



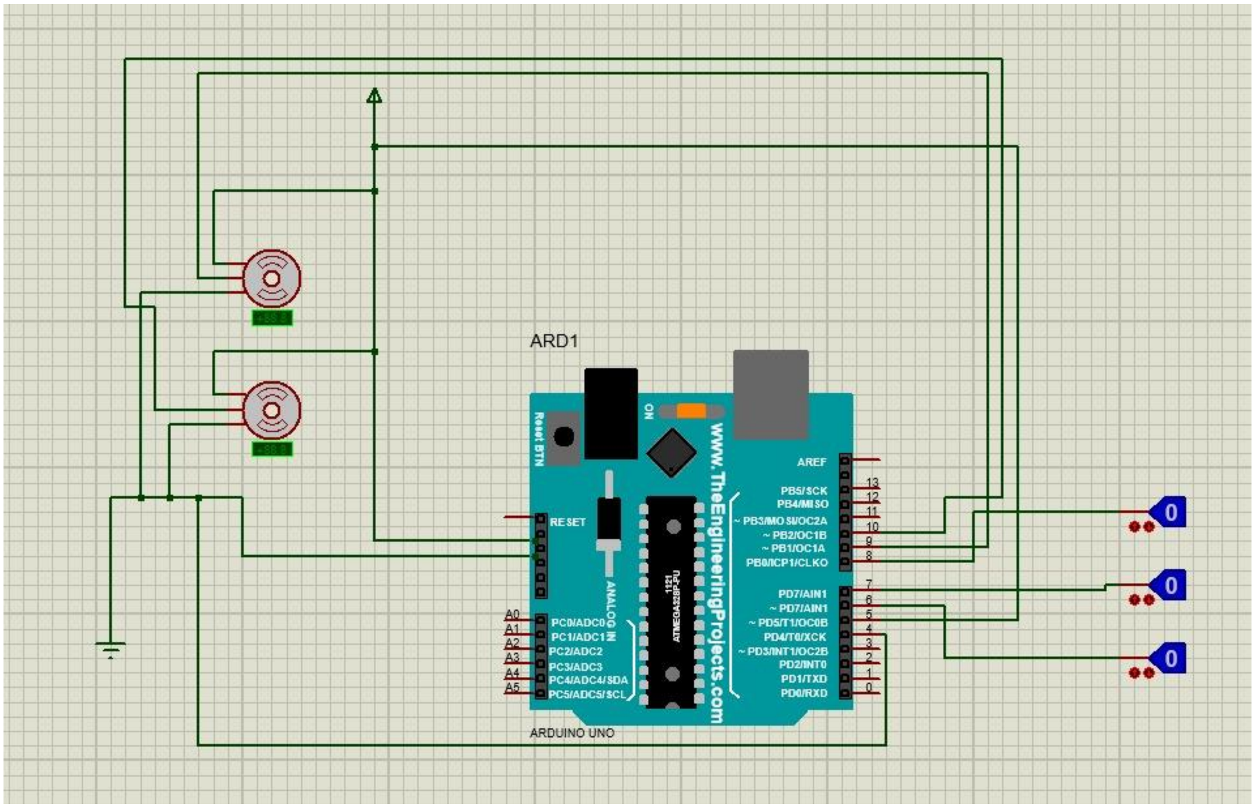


Wiring diagram

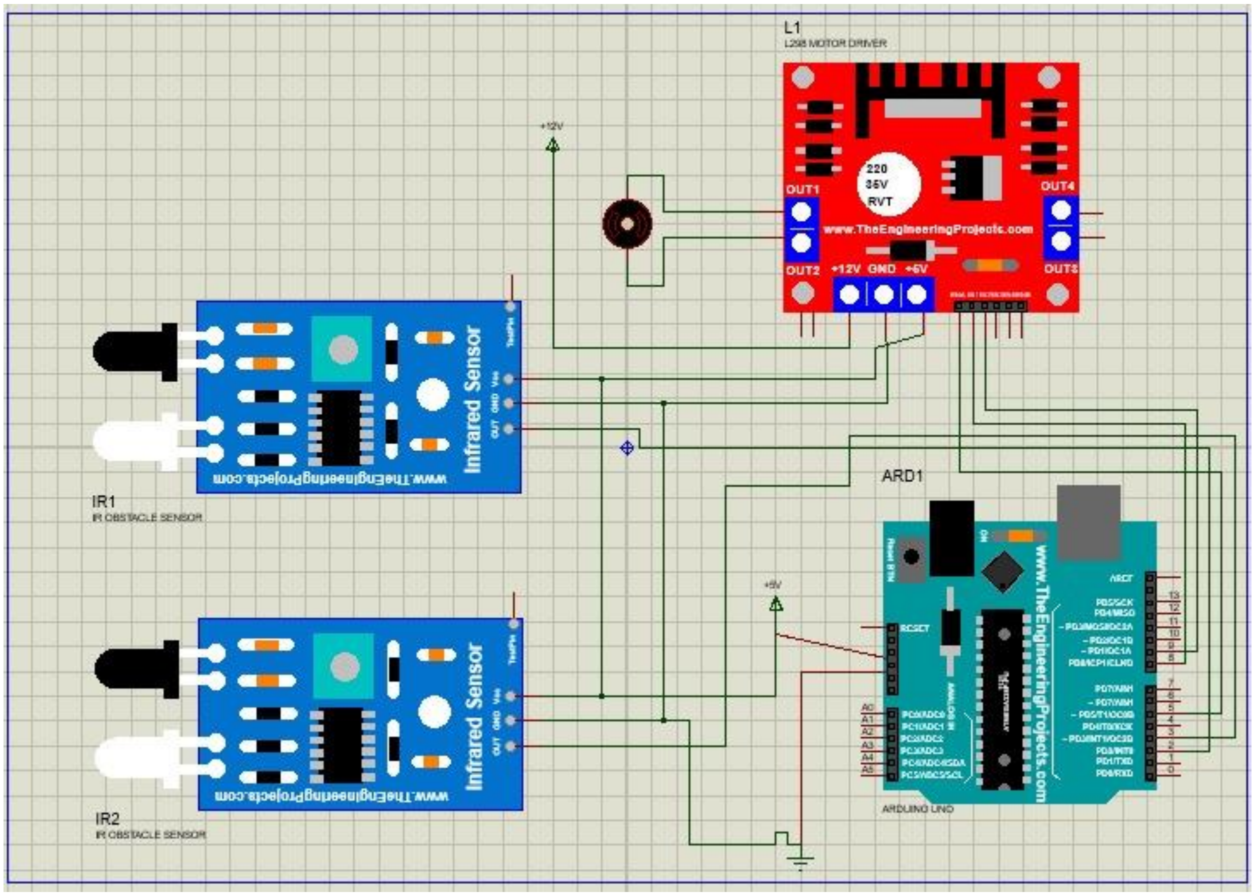
For shooting mechanism:



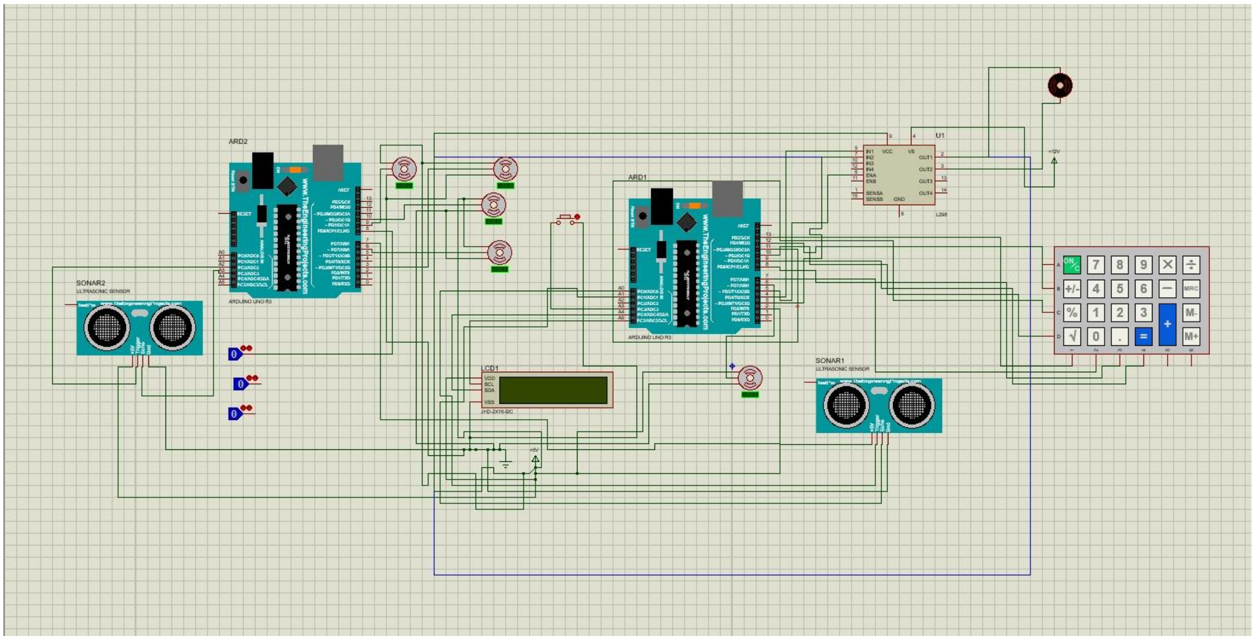
For sorting mechanism:



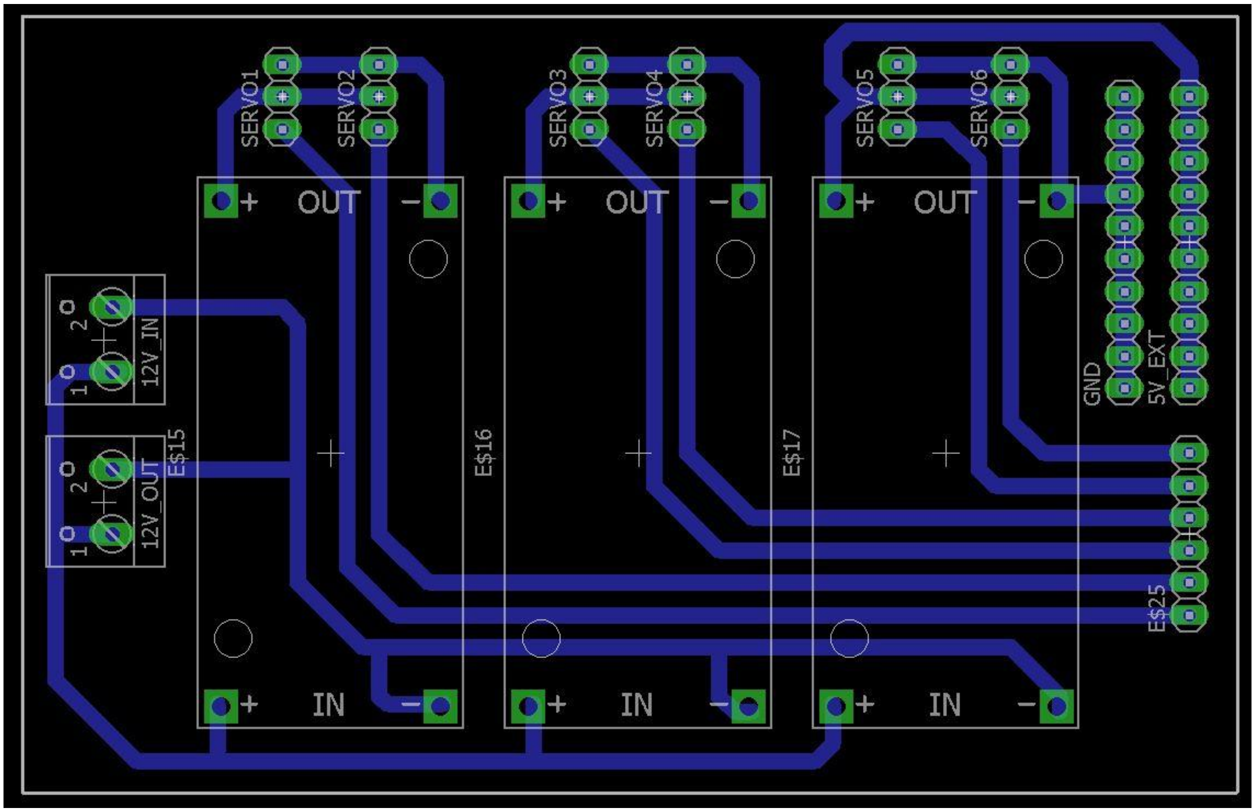
For lifting mechanism:

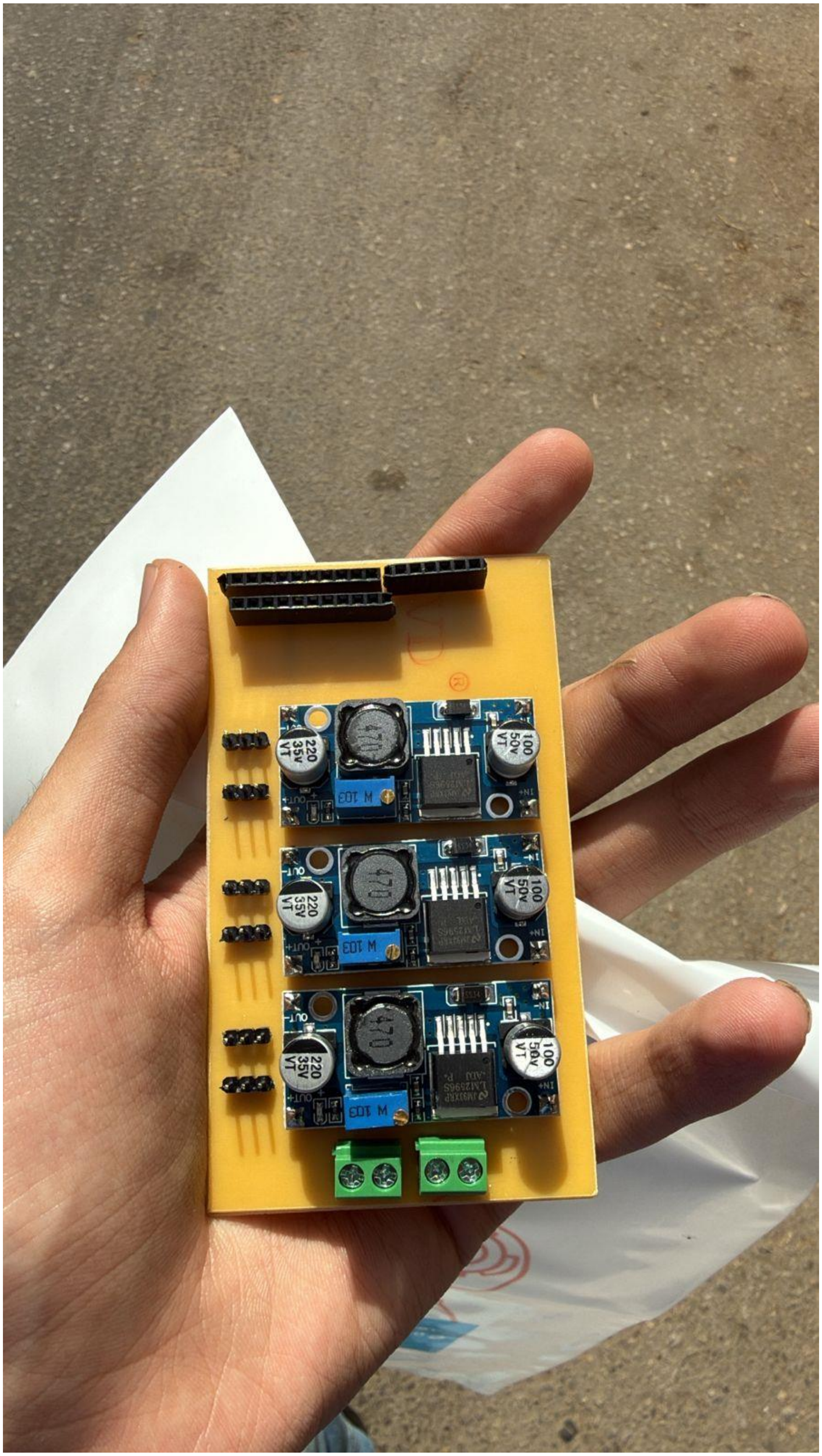


For the whole integrated system:



PCB


SERVO1
SERVO2
SERVO3
SERVO4
SERVO5
SERVO6
OUT
OUT
OUT
IN
IN
IN
12V_IN
12V_OUT
E$15
E$16
E$17
E$25
GND
5V_EXT


470
220
35v
VT
100
50v
VT